

ROBOT WORLD MODELS ARE NOT INVARIANT TO HOW THE ACTIONS ARE WRITTEN

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ABSTRACT

A robot policy is trained with one of two action parameterizations: absolute joint targets, or deltas relative to the current state. The choice is a live engineering decision in robot learning, and a world model conditioned on actions inherits it silently. We show that the inheritance is catastrophic. A latent dynamics model trained on one parameterization and handed the *identical commanded trajectory* written in the other collapses: retrieval degrades by 2.6–13.4 $\times$ across three robot datasets and two morphologies, goal-conditioned action selection falls from 53% to 15%, and on PushT the two beliefs about the same future are near-orthogonal ($\cos = 0.067$, worst case -0.377), so the predictor does not degrade gracefully, it answers a different question. This is not a distribution-shift artifact in the usual sense: the two encodings are mutually reconstructible at $R^2 = 0.996$ given the joint input, so no information is lost in the change, and we give the test that establishes this and separates a valid re-parameterization from a lossy summary or a sensor swap. The test rejected three of the four axes we proposed. The defect lives in the *action* channel, which the invariance literature for visual models does not examine: work there concerns crops, jitter and camera pose, while the parameterization of the commands themselves goes unaudited. The repair is averaging over the two encodings, and where it goes matters. Averaging the objective restores task performance by itself, and we report that rather than crediting it to anything of ours; averaging the *outputs*, which is safe for probabilities by concavity, is not available for direction-valued prediction, where we show the normalized mean can score below every member of the orbit. What objective-averaging leaves behind is the tail: worst-case agreement stays at 0.78, a disagreement penalty closes it to 0.995, and over a latent rollout it is the difference between a worst case that erodes and one that holds. On PushT, averaging alone does not repair the axis at all.

1 INTRODUCTION

A robot policy has to commit to how it writes an action. Position control emits absolute joint targets; delta control emits offsets from the current state. Both are standard, both appear in widely used datasets and codebases, and for a given task the choice is made on grounds that have nothing to do with learning dynamics: the controller’s interface, the teleoperation rig, what the previous project used. The two describe the same commanded trajectory, and given the current state s_t each determines the other exactly, since $a_{\text{rel}} = a_{\text{abs}} - s_t$.

An action-conditioned world model inherits that choice without ever being told it was made. It receives (z_t, a) and predicts the future, and nothing in its training signal indicates that a is a *notation* for a trajectory rather than the trajectory itself. This paper asks what the inheritance costs, and the answer is that it is close to total. A predictor trained on one parameterization and handed the identical commanded trajectory in the other collapses:

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retrieval degrades by up to $13.4\times$, goal-conditioned action selection falls from 53% to 15%, and on one dataset the two beliefs about the same future are near-orthogonal. On its worst input the predictor’s two answers point in opposite directions.

Why this is not the usual brittleness story. Two things separate it. The information content is provably unchanged: given the joint input the encodings reconstruct each other at $R^2 = 0.996$, so there is no sense in which the second presentation tells the model less, and degradation is not the correct response to it. And the failure is not graceful. A model pushed off-distribution normally gets noisier; this one answers a different question. The distinction matters because it decides whether the behaviour should be repaired or accepted, and §3 gives the test that makes it, which also rejected three of the four candidate axes we proposed before this one.

The channel nobody audits. Work on invariance in visual models is, almost without exception, about the pixel side: crops, colour jitter, camera pose, augmentation policies. That is where robustness is measured and where equivariant architectures are built (Cohen and Welling, 2016; Satorras et al., 2021). The action channel is left alone, and it is the channel the robot is actually commanded through.

Our setting isolates it. World models differ in how much of the learning signal comes from pixels: the Dreamer line reconstructs observations as part of its objective (Hafner et al., 2023), while decoder-free and joint-embedding models do not (Hansen et al., 2024; Assran et al., 2023; 2025; LeCun, 2022). We are in the second camp, and that is not itself a contribution, since it describes a good deal of recent work. What it buys is diagnostic. When a reconstruction term is present, the pixel signal constrains the latent directly and the action channel is not carrying the conditioning alone, so an action-side defect is partly masked by supervision from elsewhere. Here the encoder is frozen, never inverted, and never updated: the only thing the predictor learns is how actions move latents. Anything we measure is therefore a property of the transition data rather than of a rendering, an augmentation policy, or an encoder trained alongside the defect.

Contributions.

- **A failure mode in action-conditioned world models (§5.1).** Absolute versus relative action parameterization, a choice made for controller-interface reasons, determines whether a learned dynamics model can use the commands at all. Three robot datasets, two morphologies, and on PushT the two beliefs about one trajectory are near-orthogonal rather than merely degraded.
- **A test that separates a re-parameterization from a lossy one (§3).** Predictive parity is necessary, and Theorem 2 prices its failure exactly; mutual reconstructibility is a sufficient certificate rather than a gate; and both must be evaluated on the joint input, since the action channel alone reports a confident false negative ($R^2 = 0.757$ against 0.996). The test rejected three of the four axes we proposed, which is what makes the fourth worth reporting.
- **The repair, and an honest account of what each part buys (§4).** Averaging the objective over the two encodings restores task performance on its own. Averaging the *outputs* does not transfer from the probability case: we show by construction that the normalized orbit mean of direction-valued predictions can score below every member of the orbit (Proposition 3). A disagreement penalty adds no average accuracy over objective-averaging, and we say so; what it supplies is the tail, $0.78 \rightarrow 0.995$ worst case, and a worst case that holds rather than erodes over a rollout.
- **Evidence in the transition data, not the pixels (§5).** The encoder is frozen and never inverted, so nothing here is an augmentation or rendering effect: the defect is in the states and actions.

2 WHY A PREDICTOR HAS NO REASON TO RESPECT A NOTATION

Nothing in the training objective distinguishes a notation from a fact. A model fit by log loss on (z_t, a_{abs}) pairs will use whatever in a_{abs} predicts the future, including the part that is an artifact of writing commands as absolute targets rather than deltas. If that artifact correlates with the answer on the training distribution, using it shortens the description, and the optimizer does what it was asked to do.

That intuition has an exact price. Write Z for what the two encodings share and G for the coordinate that distinguishes them, and let $I_{\hat{P}}^{\mathcal{F}}(Y; G \mid Z)$ be the per-sample code length a predictor class gives up by being restricted to predictors that cannot see G . Under two-part MDL the comparison between the restricted and unrestricted class is

$$\text{MDL}_n(\mathcal{F}_{\text{inv}}) - \text{MDL}_n(\mathcal{F}) = \underbrace{L(\mathcal{F}_{\text{inv}}) - L(\mathcal{F})}_{\text{architectural}} + n \cdot I_{\hat{P}}^{\mathcal{F}}(Y; G \mid Z), \quad (1)$$

so ignoring the notation is preferred only while the architectural saving covers the per-sample term (Appendix E). Two consequences matter here. The per-sample term is what decides whether more data helps, and it takes three distinct values depending on the transformation, which is why §3 has to establish which case applies before any measurement means anything. And the architectural term is the reason a representation that already discharges the distinction costs nothing to make invariant: an encoder that never sees absolute joint targets cannot prefer them. Our axis is the case where no such representation is available, since both parameterizations are natural and neither is canonical.

Definition 1 (Make/break criterion). Let M be a task metric and A an orbit-dispersion measure. Compute bootstrap confidence intervals for ΔM and ΔA on held-out data. Report MAKE if the interval for ΔA is strictly negative and the interval for ΔM is bounded below by $-\varepsilon$; otherwise BREAK.

Orbit-constancy always reduces dispersion but need not preserve accuracy, and a drop is the criterion rejecting the proposed invariance.

3 IS IT A RE-PARAMETERIZATION, OR IS IT A DIFFERENT INPUT?

Enforcing orbit-constancy over a transformation that relabels the input is a projection onto invariants; doing it over one that changes *what is known* is blurring. The two are indistinguishable from the resulting number, so a dispersion measurement is interpretable only alongside a verdict on the transformation that produced it.

Both properties we test follow from one observation. A candidate symmetry arrives as a family of *views* $\varphi_1, \dots, \varphi_K$ carrying the situation X into a common presentation space, the predictor receiving one and not told which. It is orbit-constant exactly when $f(\varphi_v(x)) = f(\varphi_w(x))$ for all x, v, w , that is, when it is measurable with respect to the quotient of that space by the relation those equalities generate. Write Q for the class of $\varphi_v(X)$ under that relation, which does not depend on v , and $R(\varphi_v) := H(Y \mid \varphi_v(X))$ for the best risk from view v alone.

Theorem 2 (Parity is necessary, and the parity gap is its price). *Let R_{inv} be the best log loss achievable by an orbit-constant predictor. Then $R_{\text{inv}} = H(Y \mid Q)$ and*

$$R_{\text{inv}} \geq \max_v R(\varphi_v), \quad \text{so} \quad R_{\text{inv}} - \min_v R(\varphi_v) \geq \delta := \max_v R(\varphi_v) - \min_v R(\varphi_v).$$

The proof is three lines (Appendix F): Q is a function of $\varphi_v(X)$ for *every* v , so conditioning on it can only raise risk, and orbit-constancy is therefore at least as costly as the worst view in the family. The consequences are the test.

Predictive parity (necessary). Used in isolation the views must be equally predictive of y , and Theorem 2 says what a failure costs: the parity gap δ is a lower bound on what enforcing orbit-constancy gives up against simply using the better view. This is why *channel choices* and *lossy summaries* must be rejected rather than repaired. It also prices the axes we discarded: the two cameras differ by 6.6 against 23.3 mean rank, so enforcing constancy across them buys agreement by discarding the better sensor.

Reconstructibility (sufficient, not necessary). If each view is recoverable from the other then Q generates the same σ -algebra as each $\varphi_v(X)$, so $R_{\text{inv}} = R(\varphi_v)$ and orbit-constancy is free: bijective relabelling is costless, which is the certificate. It is not necessary. A *quotient* symmetry discards a nuisance coordinate by construction, and from $\theta + 2\pi k$ one cannot recover k , yet $\delta = 0$ because k says nothing about y , so Theorem 2 still gives zero cost. Requiring bijectivity would reject valid symmetries, so we report it as a certificate rather than a gate.

Evaluate on the joint input. Admissibility is a property of what the predictor receives, not of one channel. Absolute versus relative action encoding *fails* reconstruction on the action channel alone ($R^2 = 0.757$) and *passes* jointly with the state ($R^2 = 0.996$), because $a_{\text{rel}} = a_{\text{abs}} - s_t$ is a bijection only given s_t . Testing a channel in isolation produces a confident false negative.

In our own use the test rejected three transformations we had proposed: photometric jitter (an augmentation, not a group action), camera choice (different sensors carry different information: one view reached rank 6.6 where another reached 23.3), and action-window summaries (`first/mean/mean_std/concat` form a lossy hierarchy, task R^2 0.986 versus 0.709). The test also partitions our own experimental settings, six of which are group-structured and receive a certificate while the four action-conditioning settings are *action ablations* rather than orbit-constancy tests: action-shuffle permutes actions across examples, generates no orbit, and so has no dispersion measure for Definition 1 to consult (Appendix B). That leaves action-conditioned dynamics without an orbit-constancy test at all, which is where the experiments begin.

4 REPAIR: AVERAGING OVER THE TWO ENCODINGS

The repair is orbit averaging. Everything before this section exists to establish what one is entitled to average over, because averaging is a single line and the entitlement is not. What remains is a choice of *where* the average goes, and the output space decides it.

Given a certified axis, let $\{\phi_v\}$ enumerate the equivalent presentations. We train a single predictor g_θ to fit the target under every presentation while agreeing with itself across them:

$$\mathcal{L}(\theta) = \frac{1}{K} \sum_v \ell(g_\theta(z, \phi_v(a)), y) + \lambda \cdot \frac{2}{K(K-1)} \sum_{v < w} d(g_\theta(z, \phi_v(a)), g_\theta(z, \phi_w(a))), \quad (2)$$

with ℓ the task loss (InfoNCE against the true future for retrieval targets, squared error for regression) and d a disagreement penalty. The first term is an average over the orbit, so this *is* orbit averaging; what has moved is its position. The average is taken over the loss rather than over the predictions, and the predictor is never asked to match a summary of its own outputs. It is asked to be right under every presentation, and to be consistent across them.

Where the average belongs. The natural alternative puts it on the outputs: form the orbit mean of a realization-sensitive teacher’s predictions and fit a student to it. Where the prediction is a probability that is not merely reasonable but guaranteed, writing $\bar{p} := \frac{1}{K} \sum_k f(\cdot | g_k \cdot x)$, the empirical Jensen gain

$$\hat{J}(x; y) := \frac{1}{K} \sum_{k=1}^K (-\log f(y | g_k \cdot x)) - (-\log \bar{p}(y)) \geq 0$$

is non-negative by concavity of the logarithm (Appendix G), so the mixture’s log loss is never worse than the average of its members’. Averaging the outputs is therefore the right move whenever the prediction is a probability, and the point of what follows is that our setting is not that case.

The guarantee is a property of the output space, not of the symmetry, and it does not survive the move to a direction in a learned representation. There the target is a unit vector, the discrepancy is an angle, and the mean of unit vectors that disagree in direction is short and points at the centroid, so renormalising it need not return a member of the orbit. The failure is not a corner case.

Proposition 3 (No Jensen guarantee for direction-valued prediction). *Let predictions and target lie on S^{d-1} with score $\cos(\bar{u}, y)$. There exist a finite orbit and predictions $\{u_k\}$ for which the normalized mean $\bar{u} = \sum_k u_k / \|\sum_k u_k\|$ scores strictly below every u_k , and others for which $\sum_k u_k = 0$ and $\bar{u}$ is undefined.*

A witness for the first is $d = 2$, $y = (1, 0)$ and $u_1 = (1, 0)$, $u_{2,3} = (-0.9, \pm\sqrt{1-0.81})$: the members score 1, -0.9 , -0.9 while $\bar{u} = (-1, 0)$ scores -1 . For the second, take $u_{1,2} = (0, \pm 1)$. The first witness is stable, since the inequality is strict, so no genericity assumption removes it; the second is not, but $\bar{u}$ is ill-conditioned throughout a neighbourhood of it (Appendix G). This is why output averaging is the right move wherever the prediction is a probability and fails here: Table 2 shows the orbit-mean student at rank 11.55 against 5.25 for Eq. (2). Eq. (2) sidesteps the issue by never forming the mean. The predictor is asked to fit the target under each presentation and to agree with itself, which is well defined whatever space the prediction lives in.

5 EXPERIMENTS

§3 left action-conditioned dynamics without an orbit-constancy test. We supply one, measure it on three robot datasets across two morphologies, and close with what decides whether such a gap exists at all.

5.1 ABSOLUTE VERSUS RELATIVE ACTION ENCODING

We predict future JEPA latents from current latents and an action window, $\hat{z}_{t+\Delta} = g_\theta(z_t, \phi(a_{t:t+\Delta}))$, with a frozen I-JEPA encoder (Assran et al., 2023) and no pixel prediction (LeCun, 2022). The axis is the *encoding of the action window*:

$$A : a_{t:t+\Delta} \text{ (absolute joint targets)} \quad B : a_{t:t+\Delta} - s_t \text{ (relative to current state).}$$

Both describe the same commanded trajectory, and position control versus delta control is a live design choice in robot policy learning, so the axis is not contrived. It is admissible: evaluated on the joint input as §3 requires, the two encodings are mutually reconstructible at $R^2 = 0.996$ with a task- R^2 gap of 0.003. Reconstruction from the action channel alone reaches only $R^2 = 0.757$, which is the false negative that section warns about.

Holding the camera fixed keeps all predictions in one latent space, so dispersion is well defined. A multi-camera axis would not be, since predictions would live in camera-specific spaces. Following Definition 1, the teacher is *realization-sensitive*: trained on the canonical encoding alone, then evaluated across the orbit.

Table 1 reports the measurement: the identical action window in the equivalent encoding collapses retrieval by 2.6–13.4 $\times$. The two ALOHA datasets agree to three decimals in cross-encoding cosine, so the effect is not task-specific, and on PushT, a different robot with 2-dimensional actions, the two beliefs about the same future are close to orthogonal. Figure 1 shows it on frames: handed the same trajectory in the other notation, the predictor retrieves a visibly different scene.

A certified special case of covariate shift. We should be exact about what this is. The protocol trains on one encoding and tests on the other, so it is a distribution change, and we claim no new phenomenon. What the certificate adds is the thing covariate-shift results normally cannot supply. Such results confound *different distribution* with *less information*, and the two have opposite implications: if the shifted input carries less about the target, degradation is correct behaviour and there is nothing to repair. Admissibility excludes that branch by construction ($R^2 = 0.996$ jointly, task- R^2 matched to 0.003), turning a claim about brittleness into a claim about brittleness *to a change that provably carries no information about the target*. Whether repair is then free in practice is a separate question, and §4 finds that it is not on the dataset where the collapse is largest. The magnitude is what makes it worth acting on: on PushT the cosine is 0.067 and the worst case is negative, so the predictor does not degrade gracefully, it answers a different question. The symmetry is one of the data pipeline rather than of the world, with no conservation law behind it, which is why it had to be certified rather than assumed.

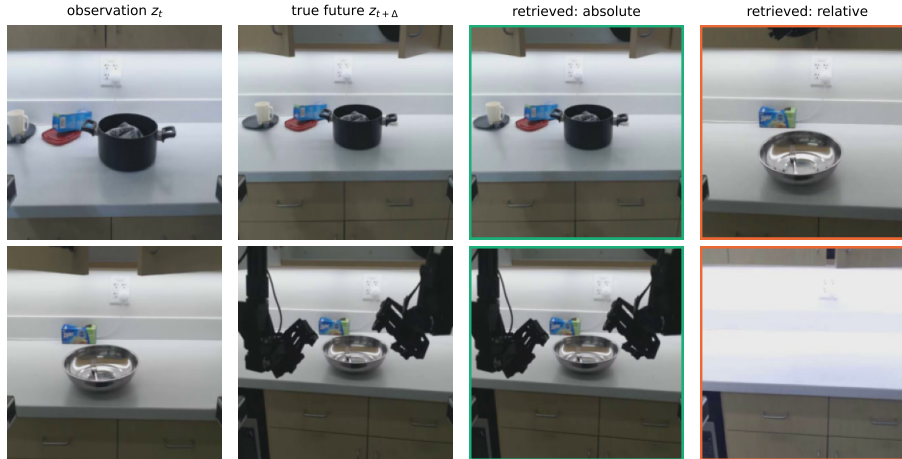


Figure 1: The same failure on frames. Each row is one test instance: the observation at t , the true frame at $t + \Delta$, and the frames retrieved when the predictor is handed the identical action window in absolute and in relative encoding. Green marks a correct retrieval, orange an incorrect one. Rows are restricted to misses from a *different* episode, since a same-episode miss is a near-duplicate frame and would show the reader nothing. Of 384 test instances, 125 are absolute-correct and relative-wrong, 95 of them from a different episode; two are shown.

Table 1: Orbit dispersion under the action-encoding axis. $\Delta = 4s$, 3 seeds, episode-disjoint splits, pool size 384. Collapse is held-out rank over canonical rank; cos is agreement between the two encodings’ predictions, mean and worst case. ALOHA datasets (Zhao et al., 2023); PushT (Chi et al., 2023); all via Cadene et al. (2024).

Dataset	Morphology	Retrieval rank			cos	
		Canon.	Held out	Collapse	mean	worst
aloha_mobile_cabinet	14-D bimanual, 50 fps	6.19	61.5	9.9×	0.370	−0.178
aloha_static_coffee	14-D bimanual, new task	3.97	53.3	13.4×	0.371	−0.096
pusht	2-D planar pusher, 10 fps	66.2	169.8	2.6×	0.067	−0.377

The failure is a control failure. Retrieval rank measures representation quality; control asks which action reaches a goal, scoring candidates under the learned dynamics in the same encoding as the query. The teacher selects the correct action 53.1% of the time in its own encoding and 15.4% in the equivalent one (Table 2), and the mean cosine of 0.370 understates the tail, whose worst case is -0.178 . A model that cannot choose an action when the same commands are written differently is not usable for control, whatever its retrieval score.

Repair, and what each part of it buys. Table 2 separates three things that are easy to conflate, and the separation is not flattering to the penalty, so we state it first. *Averaging the objective over the orbit recovers the task on its own.* Training on both encodings with the penalty off ($\lambda = 0$) takes held-out rank from 61.5 to 5.98 and held-out action selection from 15.4% to 40.1%; $\lambda = 16$ moves those to 5.19 and 41.1%, one point against a seed standard deviation of four. On this evidence **the penalty does not improve average task performance over averaging alone**, and we do not claim that it does.

What it buys is invariance in the sense the word is usually meant. At $\lambda = 0$ the worst-case agreement across the orbit is 0.779: on average the two notations agree, and on the inputs where they do not the predictions are still far apart. The penalty closes that, $0.779 \rightarrow 0.995$, a four-fold reduction in worst-case disagreement and well outside seed noise. The price is about three points of canonical action selection, $44.3\% \rightarrow 41.1\%$. So the honest summary is that once the axis is known, augmentation restores accuracy and the penalty converts “usually agrees” into a guarantee, which is what Theorem 2 is stated in terms of and what a

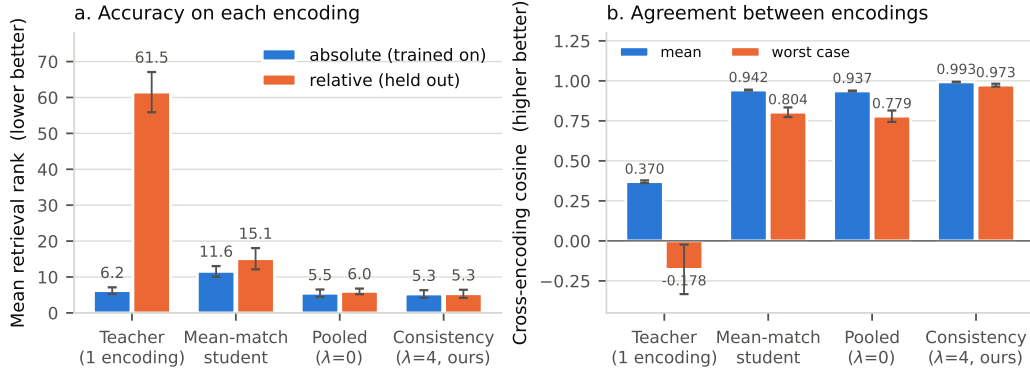


Figure 2: Left: retrieval rank on each encoding. Right: cross-encoding agreement, mean and worst case over the evaluation set. The teacher is competent on the encoding it was trained on and collapses on the equivalent one. Pooled training ($\lambda = 0$) is shown because it, not mean-matching, is the baseline that competes: it recovers accuracy but leaves worst-case agreement at 0.779, which only the consistency term lifts. The teacher’s worst case is -0.178 : on its worst input the two notations point in opposite directions.

Table 2: Repair on `aloha_mobile_cabinet`, 3 seeds, all rows from the same runs. λ is the disagreement weight of Eq. (2); $\lambda = 0$ is the same two-view pooled training with the penalty off, which is the baseline any practitioner would reach for once the axis is known; the orbit-mean row fits a student to the averaged teacher prediction instead. cos is mean and worst-case pairwise agreement across the orbit. Action selection is top-1 over 32 candidates, chance 0.031.

	Retrieval rank		Agreement		Action selection	
	Canon.	Held out	mean	worst	Canon.	Held out
Teacher	6.19	61.5	0.370	—	0.531	0.154
Orbit-mean student	11.55	15.1	0.942	0.804	—	—
$\lambda = 0$	5.46	5.98	0.937	0.779	0.443	0.401
$\lambda = 1$	5.51	5.58	0.981	0.939	0.408	0.406
$\lambda = 4$	5.25	5.31	0.993	0.973	0.426	0.402
$\lambda = 16$	5.17	5.19	0.999	0.995	0.411	0.411

certificate needs. Fitting the orbit mean does neither: it reaches 0.942 mean agreement at roughly twice the rank error, for the reason Proposition 3 gives.

The mean is stable over a horizon; the worst case is not. Every number above is one step, and a world model is used over many. We roll the predictor forward on held-out episodes for hops of 4s each, feeding its own output back in with the true action window and state supplied at every hop, so the only iterated quantity is the model’s belief (Table 3). This is not closed-loop control, and predictive quality decays over the horizon (cosine to the true latent $0.33 \rightarrow 0.13$ on cabinet), so later hops describe agreement about an increasingly uncertain quantity.

On the two ALOHA datasets mean agreement is close to flat for every model, so the disagreement does not compound in the average and we make no such claim. Worst-case agreement behaves differently: the teacher erodes, pooled training slows that without stopping it, and only the consistency term holds. PushT separates the two far more sharply, because there *pooled training does not repair the axis at all*: mean agreement reaches 0.633 and worst case is *negative*, -0.348 by the second hop, so a predictor trained on both encodings still produces opposed beliefs about the same commanded trajectory. The penalty lifts those to

Table 3: Latent rollout on held-out episodes, 3 seeds, three datasets and two morphologies: cross-encoding cosine at the final hop (three hops of 4s, two for PushT, whose episodes are short). Mean and worst case over the evaluation set. Per-hop trajectories are in Table 4. Chains need horizon $\times \Delta$ of runway and so start earlier in an episode than the pairs of Table 2, so read the comparison across models, not against that table.

Dataset	Teacher		$\lambda = 0$		$\lambda = 16$	
	mean	worst	mean	worst	mean	worst
ALOHA cabinet	0.805	0.242	0.988	0.896	1.000	0.997
ALOHA coffee	0.898	0.614	0.991	0.955	1.000	0.998
PushT	0.178	-0.497	0.677	-0.348	0.872	0.139

0.946 and +0.139 without closing the gap. The easier the case, the more averaging alone suffices; the harder it is, the more of the work the consistency term does.

On the value of finding the axis. That averaging over the orbit works, once you know which orbit, is not a weakness of the diagnosis; it is the point of it. The averaging is one term in Eq. (2). What it presupposes is an admissible transformation to average over, and three of the four we proposed were not one: had we averaged over them, the cost would have been the parity gap of Theorem 2, paid for nothing. The entitlement is the hard part, which is why it occupies §3 and this section is short.

Where repair is not free. On PushT the repaired predictor improves worst-case rank $169.8 \rightarrow 85.7$ and $\cos 0.067 \rightarrow 0.971$, but costs canonical accuracy ($66.2 \rightarrow 79.2$). That teacher is weak in absolute terms (rank 66 of 384), and §5.2 shows the repair is free only where the model has slack.

5.2 THE DEFECT IS FLAT IN n OVER THE RANGE WE CAN SWEEP

The action-encoding axis is regime (iii) of §2: the teacher is trained on the canonical encoding only, so the coordinate distinguishing the two notations is never exercised by a label. Equation (1) therefore prices nothing along it, and the prediction to test is that the departure does not close as the model gets better at its own task. Sweeping n by slicing a fixed cached training set, so that latents are identical across conditions, the collapse ratio grows monotonically: $2.5\times, 3.9\times, 5.4\times, 9.9\times$ for $n = 150, 300, 600, 1200$ (Figure 3).

The ratio alone would overstate this, and we read it carefully. Cross-encoding cosine is *flat* in n (0.357, 0.337, 0.374, 0.370): more data does not make the predictor disagree with itself more. It makes the same disagreement more costly, because the canonical model sharpens while the held-out one does not, and the ratio is a quotient of the two. The claim the sweep supports is therefore the weaker and more useful one, that the defect is invariant to scale rather than amplified by it: at $n = 1200$ the predictor is eight times better on the encoding it saw and no better at all on the one it did not. The same pattern holds for optimization and for capacity (Table 5), with the same caveat and one more. The larger and longer-trained teachers are also worse on their own encoding, so both are partly overfitting 1200 examples; the cosine comparison is the defensible half, and a clean capacity claim would require matched canonical accuracy, which we do not have. We do not read the capacity column as independent evidence.

Locating $I(Y; G | Z)$. Sweeping λ (Figure 4) exposes the cost term directly. At $n = 150$ a strong penalty ($\lambda = 256$) is worth -6.59 rank; at $n = 1200$ it costs $+0.38$. The sign flips as n grows, as the right-hand side of Eq. (1) predicts. Below the crossover orbit-constancy is free regularization; above it, it is a real cost. The useful operating point, $\lambda \in [4, 16]$, lies below.

5.3 WHY THIS AXIS IS THE HARD CASE

Whether a re-parameterization costs anything is decided by the representation, not by the transformation. Two control axes with known analytic structure (Appendix D.6) show dispersion at 10^{-15} and 10^{-12} , which is to say none, and the reason is not robustness: the Kepler featurization encodes an angle as $(\sin M, \cos M)$, so M and $M + 2\pi k$ are literally the same input, and the Lorenz predictor never receives absolute time. The transformation never reaches the model. A molecular-dynamics ablation makes the same point with the variable isolated (Appendix H.2): same system, same data, same criterion, and the verdict flips from MAKE to BREAK purely by exchanging an invariant encoder for raw coordinates.

This is the architectural term $L(\mathcal{F}) - L(\mathcal{F}_{\text{inv}})$ of Eq. (1) made visible: **a gap exists exactly when the representation does not already discharge the distinction**. It is also why absolute versus relative action encoding is the hard case rather than a curiosity. For a camera angle or a time origin one can featurize the nuisance away, and should. Here both parameterizations are natural, neither is canonical, and a predictor that takes an action window as a vector of numbers has no way to know which convention produced it. There is no featurization to reach for, which leaves measuring the gap and training it away.

6 DISCUSSION

Discovering rather than assuming invariance. The question of which transformations a predictor should be invariant to is the one invariant risk minimization (Arjovsky et al., 2019) and the causal-invariance line (Peters et al., 2016) also ask, and the contrast is informative. Those methods are handed a partition into environments and search for a representation whose optimal predictor is stable across it, so invariance is *inferred* from variation the data already contains. We are handed a candidate transformation and asked whether enforcing constancy over it is projection or blurring, which is decidable from a single environment and is a property of the transformation rather than of an environment family. The two are complementary, and neither subsumes the other: environments say an invariance exists without naming the group, certification says whether a named group is one. Proposing transformations automatically and certifying them with Theorem 2 is the obvious thing to build next.

Relation to prior work. Building invariance into the hypothesis class (Cohen and Welling, 2016; Satorras et al., 2021) is the left-hand side of Eq. (1) and is preferable wherever it applies, since the price is zero; our concern is what it does not cover. The generalization benefits are established (Elesedy and Zaidi, 2021) and augmentation reads as approximate marginalization over a nuisance group (Chen et al., 2020), which Eq. (1) prices. Latent-space world models (Assran et al., 2023; 2025; LeCun, 2022) are usually evaluated with representation-space metrics and an action-channel ablation, and comparison with reinforcement-learned ones (Hafner et al., 2023; Hansen et al., 2024) is orthogonal: we claim not better forecasting but that a forecast should not depend on which of two equivalent notations was handed over. We release Mezzanine, which implements the procedure end to end.

Where repair is unavailable. The repair needs a certified symmetry whose orbit can be enumerated, and both conditions bind. Where an orbit is large or continuous, enumerating it is out of reach and no post-hoc repair is available at any price (Appendix D.2). Our own measurements bracket it: flattened LJ coordinates make the orbit $S_n \times SE(3)$ and enumeration cannot preserve accuracy, while the action-encoding axis is tractable precisely because it has two elements.

Limitations. The predictor is a small MLP on frozen latents and three seeds is thin, so we report directions rather than magnitudes, and the capacity ablation’s larger teacher is also worse on its own encoding, so it is not independent evidence. Evaluation is offline: action selection is a proxy for control, and the rollout is closed in the latent only. The admissibility probe is linear ridge regression, so it certifies linear relabellings trivially and may reject nonlinear bijections.

Conclusion. Averaging over the orbit is the repair, and it is the easy part. Which orbit, whether the transformation is a symmetry at all, and where the average belongs once the prediction is a direction: those come first, and they are the paper.

REPRODUCIBILITY STATEMENT

All results regenerate from the accompanying repository. Theorem 2 is proved in Appendix F and Proposition 3 in Appendix G; Appendix A gives datasets, architecture, hyperparameters, seeds and hardware. The robotics experiments use public LeRobot datasets; the exact commands, seeds and run directories are listed in the repository, latents are cached by world and encoder fingerprint, and every reported number comes from a stored `results.json`. Run-to-run variation is reported as standard deviation over three seeds throughout. We note that ranks vary by 2–4% between CPU and GPU from floating-point ordering while the dispersion measures agree to three decimals, so the compute device is recorded alongside the seed.

ETHICS STATEMENT

This work studies failure modes of predictive models and proposes a diagnostic and a repair. It uses public robot-learning datasets and involves no human subjects, no personally identifying data, and no new data collection. The diagnostic is intended to be run before a learned dynamics model is trusted for control rather than after, and we note that the repair buys worst-case agreement at a measurable cost in average accuracy, so the operating point is application-specific rather than universal.

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A EXPERIMENTAL DETAIL

Datasets. All three are public LeRobot releases (Cadene et al., 2024), used unmodified.

Dataset	Camera	Episodes	Frames	fps	Δ (steps)	train/test eps.
<code>aloha_mobile_cabinet</code>	<code>cam_high</code>	85	127,500	50	200	68 / 17
<code>aloha_static_coffee</code>	<code>cam_high</code>	50	55,000	50	200	40 / 10
<code>pusht</code>	<code>image</code>	206	25,650	10	40	165 / 41

Splits are *episode-disjoint*: no frame from a test episode appears in training, so a nearest-neighbour retrieval cannot succeed by recalling an adjacent frame of a seen trajectory. Δ is 4 seconds in every case, which is 200 steps at 50 fps and 40 at 10 fps.

Encoder. Frozen I-JEPA ViT-H/14 (Assran et al., 2023) (`facebook/ijepa_vith14.1k`), fp16, batch size 32. Frames are resized to 224 at cache-build time. Embeddings are the concatenated mean and standard deviation over patch tokens of the fourth-from-last block. The encoder is never fine-tuned, never inverted, and no gradient reaches it; latents are cached and keyed by world and encoder fingerprint together with the resize, so a run cannot silently mix provenance.

Predictor. An MLP on $[z_t \parallel \phi_v(a_{t:t+\Delta})]$ predicting the ℓ_2 -normalised future latent: hidden width 1024, depth 2, trained 800 steps with InfoNCE against the true future, in-batch negatives. The action window is 16 points sampled uniformly across the Δ horizon, so the action input is $16 \times d_a$ ($d_a = 14$ for ALOHA, 2 for PushT). Latents are centred using statistics from the training split only. $n_{\text{train}} = 1200$, $n_{\text{test}} = 384$.

Repair. Eq. (2) with $\lambda \in \{0, 1, 4, 16\}$, the same 800 steps, d the mean pairwise cosine disagreement. $\lambda = 0$ is identical two-view pooled training with the penalty switched off and is the baseline throughout.

Metrics. Retrieval rank is the position of the true future latent among the 384 test latents under cosine similarity. Cross-encoding agreement is the cosine between the two encodings’ predictions for the same input, reported as a mean and as the minimum over the evaluation set. Goal-conditioned action selection scores 32 candidate action windows under the learned dynamics against the true future and reports top-1; chance is 0.031.

Seeds and hardware. Every number is the mean over three seeds (0, 1, 2) with the standard deviation where reported; the seed controls split sampling, initialisation and batching. All runs are on a single RTX 5090. With latents cached, a full three-seed sweep on `aloha_mobile_cabinet` takes a few minutes; the first run on a dataset must decode video and takes longer.

What is not controlled. Three seeds is thin, and we report the direction of each effect rather than its magnitude. Capacity and optimisation sweeps vary one axis against a fixed 1200-example training set, so the larger and longer-trained teachers are partly overfitting; that is stated where those numbers appear and we do not treat them as independent evidence.

B ACTION ABLATIONS ARE NOT ORBIT-CONSTANCY TESTS

Action conditioning is routinely validated by *action-shuffle*: permute actions across examples and check that accuracy collapses. Four of our settings were originally run that way (ALOHA, LIBERO-10, PushT, iPhyRE), and the test of §3 says plainly what that evidence is and is not.

Action-shuffle is deliberately not evidence-preserving. It feeds the predictor an action window belonging to a different example, so the intervention destroys information rather than re-presenting it, generates no orbit, and leaves no dispersion measure A for Definition 1 to consult. Such a result is still informative: a predictor whose accuracy collapses under shuffled actions is demonstrably using the action channel rather than ignoring it, and we report those rows as *action ablations* (Appendix H.1). What they are not is evidence that the predictor’s belief is stable under a change that preserves the commands, which is a different and stronger property and the one this paper measures.

That gap is the reason §5.1 exists. Before it, action-conditioned dynamics had no orbit-constancy test at all, only ablations that confirm the channel is read.

Table 7 adds four numerical axes with known analytic structure, which serve as controls: two have a real departure that repair removes, and two have none to begin with.

Repair beyond robotics. On node permutation, a predictor trained with Eq. (2) at $\lambda = 0$ reaches dispersion 21.2 against 54.1 for one fitted to the orbit mean, with marginally lower test error. On periodic integration it makes no difference: that predictor already uses an invariant featurization and sits at 10^{-13} .

C ROLLOUT DIVERGENCE, PER HOP

Table 4: Per-hop cross-encoding cosine for Table 3: 3 seeds, three datasets and two morphologies; cross-encoding cosine between the absolute and relative rollouts at each hop. PushT episodes are short and admit two hops rather than three. Read the trend across hops, not the absolute values: chains need horizon $\times \Delta$ of runway and so start earlier in an episode than the pairs of Table 2, where the encodings are more similar at the input (cos 0.46 against 0.27).

Dataset	Model	Mean cos			Worst-case cos		
		hop 1	hop 2	hop 3	hop 1	hop 2	hop 3
ALOHA cabinet	Teacher	0.836	0.803	0.805	0.515	0.350	0.242
	$\lambda = 0$	0.992	0.990	0.988	0.972	0.951	0.896
	$\lambda = 16$	1.000	1.000	1.000	1.000	0.999	0.997
ALOHA coffee	Teacher	0.916	0.889	0.898	0.707	0.676	0.614
	$\lambda = 0$	0.995	0.993	0.991	0.987	0.973	0.955
	$\lambda = 16$	1.000	1.000	1.000	1.000	0.999	0.998
PushT	Teacher	0.246	0.178	—	−0.298	−0.497	—
	$\lambda = 0$	0.633	0.677	—	−0.054	−0.348	—
	$\lambda = 16$	0.946	0.872	—	0.666	0.139	—

D ADDITIONAL RESULTS

Material referenced from the main text.

Table 5: The incentive strengthens along all three axes the threshold implicates. Cross-encoding cosine; lower is more broken.

Axis	Varied	From	To
Data	$n = 150 \rightarrow 1200$	collapse	$2.5\times \rightarrow 9.9\times$
Optimization	$800 \rightarrow 8000$ steps	$\cos 0.370$	$\cos 0.242$
Capacity	hidden $1024/2 \rightarrow 2048/3$	$\cos 0.370$	$\cos 0.125$

D.1 SYMMETRY BREAKING ALONG DATA, OPTIMIZATION AND CAPACITY

D.2 WHEN THE ORBIT IS TOO LARGE TO ENUMERATE

Eq. (2) sums over the orbit, so it is available only when the orbit can be enumerated. Our axis has two elements, which is why the repair is cheap; it is worth being explicit about how quickly that stops being true, because it bounds where this procedure applies.

Discrete orbits grow factorially. Enforcing invariance to the order of N interchangeable inputs requires averaging over S_N , which is $N! \approx 9.3 \times 10^{157}$ terms at $N = 100$. Monte Carlo over the orbit converges as $k^{-1/2}$ (Proposition 8), so reducing an order-induced standard deviation of $\sim 10^{-1}$ to negligible costs on the order of 10^{14} forward passes for a single prediction.

Continuous orbits are not finite sums at all. A continuous group requires integration against the Haar measure rather than a sum, and no finite sample of the orbit is exact. A camera pose treated as a continuous nuisance is in this case; an action encoding with two presentations is not.

Where that leaves robot dynamics. The tractable cases are those where the presentations are few and enumerable, which covers the parameterization choices a robot stack actually makes: absolute against relative actions, one frame convention against another, a fixed set of unit conventions. The intractable ones are those where the nuisance is continuous or combinatorial, and for those the architectural route of Eq. (1), a representation that never encodes the distinction, is the only practical answer.

D.3 TRUE SYMMETRY BREAKING AT FINITE n

Table 6: True symmetry breaking at finite n . Target $Y = \mathbf{1}\{\text{median}(\mathbf{x}) > 0 \wedge \text{std}(\mathbf{x}) > 0.8 \wedge \text{range}(\mathbf{x}) > 2\}$ is exactly S_d -invariant. The unconstrained model’s orbit dispersion under random coordinate permutations (TV to the orbit mean) is large at finite n and grows with $|S_d|$, confirming that compression selects non-invariant predictors even when the target is invariant.

d	$ S_d $	n_{train}	Acc (unconstr.)	Acc (invariant)	Orbit disp.
8	4×10^4	1,000	0.858	0.979	0.141
8	4×10^4	10,000	0.942	0.993	0.059
8	4×10^4	100,000	0.968	0.995	0.024
16	2.1×10^{13}	1,000	0.756	0.974	0.229
16	2.1×10^{13}	10,000	0.828	0.991	0.163
16	2.1×10^{13}	100,000	0.918	0.994	0.062
32	2.6×10^{35}	1,000	0.708	0.978	0.277
32	2.6×10^{35}	10,000	0.757	0.991	0.224
32	2.6×10^{35}	100,000	0.810	0.993	0.151

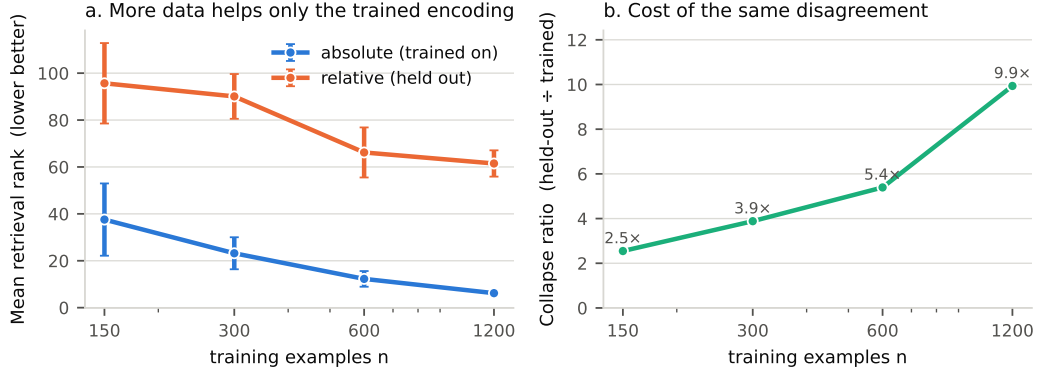


Figure 3: More data improves the trained encoding and barely improves the equivalent one, so the collapse ratio grows from 2.5 $\times$ to 9.9 $\times$. Panel b is a quotient of the two curves in panel a, and should be read as the growing cost of a fixed disagreement, not as a growing disagreement: cross-encoding cosine is flat across the same sweep (0.357, 0.337, 0.374, 0.370).

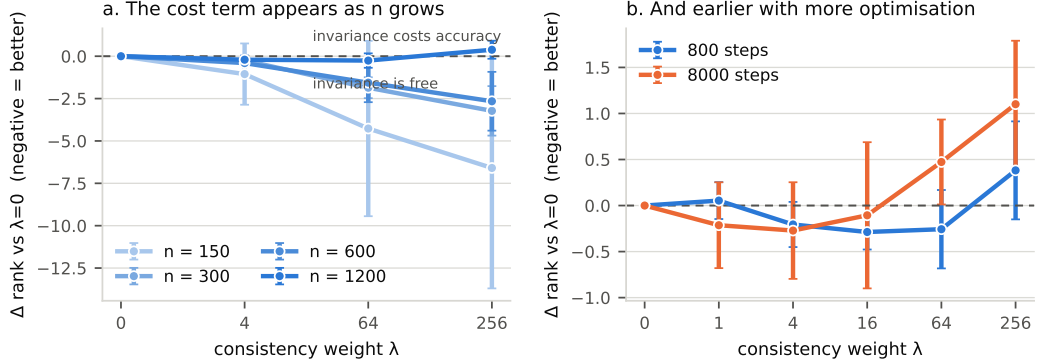


Figure 4: The cost of orbit-constancy as a function of λ , relative to $\lambda = 0$. Negative is better. Left: the benefit shrinks and finally reverses as n grows. Right: the same, earlier, with more optimization.

D.4 COLLAPSE RATIO AGAINST TRAINING-SET SIZE

Figure 3 gives the n sweep behind §5.2. Panel a is the quantity to read: more data improves the encoding the teacher was trained on and leaves the equivalent one where it was.

D.5 COST OF ORBIT-CONSTANCY VERSUS λ

Figure 4 sweeps the disagreement weight against training-set size, and locates the point at which orbit-constancy stops being free regularisation and starts costing rank.

D.6 ADMISSIBILITY CERTIFICATES ON FOUR NUMERICAL AXES

Table 7: Admissibility certificates on four further numerical axes, 3 seeds each. All are admissible; two exhibit no measurable departure because the featurization already removes the nuisance.

Axis	Family	Teacher dispersion	Reduction	Certificate
Angle wrap (Kepler)	gauge	8.6×10^{-15}	—	bijective
Time origin (Lorenz)	gauge	4.2×10^{-12}	—	bijective
Circular shift	gauge	2.6×10^{-2}	100%	quotient
Node permutation ($Ax=b$)	exchangeability	3.3×10^2	83.4%	quotient

E DESCRIPTION-LENGTH ACCOUNT OF THE INCENTIVE

This appendix collects the description-length results the main text uses, in the two-part MDL setting of Grünwald (2007) and for the strictly proper scoring rules of Gneiting and Raftery (2007). They price the incentive to exploit a notation, and they are group-agnostic: nothing below refers to which transformation is at stake, only that some coordinate of the input distinguishes presentations. Proofs are omitted where they are routine.

Throughout, G acts measurably on inputs, Z is a maximal invariant, $\hat{P}$ is the empirical law, and $\mathcal{F}_{\text{inv}} \subseteq \mathcal{F}$ is the orbit-constant subclass. The *effective orbit information* $I_{\hat{P}}^{\mathcal{F}}(Y; G | Z)$ of §2 is the per-sample code length $\mathcal{F}$ gives up by being restricted to $\mathcal{F}_{\text{inv}}$, and $\Delta_{\text{mis}} := I_{\hat{P}}^{\mathcal{F}}(Y; G | Z) - I_{\hat{P}}(Y; G | Z)$ is its misspecification component. Write $\text{Reg}_{\hat{P}}(\mathcal{F}; X)$ for the best-in-class regret against the Bayes envelope.

Three regimes, not one conclusion. Everything downstream turns on when $I_{\hat{P}}^{\mathcal{F}}(Y; G | Z)$ is positive. It is a *best-in-class* gap, so by Corollary 6 it equals $I_{\hat{P}}(Y; G | Z) + \Delta_{\text{mis}}$, within $\varepsilon_X + \varepsilon_Z$ of the population term, and three cases follow. (i) The group coordinate informs the target: the population term is positive and the penalty for invariance grows linearly in n . This is the case of an approximate or assumed symmetry, where the transformation is not one, and we note it only to set it aside; it is not the case this paper studies. (ii) The target is exactly invariant and both classes realize their infima: that term is zero, the linear growth disappears, and what remains is Δ_{mis} , a finite-sample burden growing with orbit size and decaying slowly in n (Table 6). (iii) The labels never exercise the coordinate: the objective prices it in neither direction, no reading of n applies, and the mode is simply unconstrained. §5 studies the third case. The left-hand side is architectural in all three, so **a representation that makes the invariance cheap to describe removes the incentive entirely**, which is the lever to reach for wherever an equivariant construction exists.

Definition 4 (Prequential (two-part) MDL). Given a predictor class $\mathcal{F}$ and a description-length proxy $L(\mathcal{F})$,

$$\text{MDL}_n(\mathcal{F}) := L(\mathcal{F}) + n \cdot \inf_{f \in \mathcal{F}} \mathbb{E}_{\hat{P}}[-\log f(Y | X)].$$

Theorem 5 (MDL symmetry-breaking threshold). *Under Definition 4,*

$$\text{MDL}_n(\mathcal{F}_{\text{inv}}) - \text{MDL}_n(\mathcal{F}) = \underbrace{L(\mathcal{F}_{\text{inv}}) - L(\mathcal{F})}_{\text{description-length diff.}} + n \cdot I_{\hat{P}}^{\mathcal{F}}(Y; G | Z).$$

Enforcing invariance is MDL-preferred iff $L(\mathcal{F}) - L(\mathcal{F}_{\text{inv}}) \geq n \cdot I_{\hat{P}}^{\mathcal{F}}(Y; G | Z)$.

The decomposition holds for any strictly proper scoring rule, since propriety makes functional invariance and distribution invariance coincide.

Corollary 6 (No sign flip; Bayes-envelope approximation). *Since $I_{\hat{P}}^{\mathcal{F}}(Y; G | Z) \geq 0$, necessarily $\Delta_{\text{mis}} \geq -I_{\hat{P}}(Y; G | Z)$: misspecification can cancel the conditional mutual information term but cannot reverse the direction of the best-in-class incentive. If $\text{Reg}_{\hat{P}}(\mathcal{F}; X) \leq \varepsilon_X$ and $\text{Reg}_{\hat{P}}(\mathcal{F}_{\text{inv}}; Z) \leq \varepsilon_Z$, then $|I_{\hat{P}}^{\mathcal{F}}(Y; G | Z) - I_{\hat{P}}(Y; G | Z)| \leq \varepsilon_X + \varepsilon_Z$.*

This is what makes the three regimes of §2 exhaustive: the best-in-class gap is the population conditional mutual information plus a misspecification term that is bounded but need not vanish.

E.1 ORBIT-CONSTANCY

Theorem 7 (Optimal prediction is constant on symmetry orbits). *If $p^*(y | x)$ is G -invariant and f^* is the Bayes-optimal predictor under any strictly proper scoring rule, then $f^*(y | x) = f^*(y | g \cdot x)$ for all $g \in G$, for all x, y .*

This is the licence for treating orbit dispersion as an error measure at all: under a genuine symmetry, any dispersion is a departure from the optimum. It is also why certification (§3) is load bearing, since the theorem says nothing when p^* is not G -invariant.

Proposition 8 (Monte Carlo accuracy of orbit averaging). *Define $U := f(y | g \cdot x)$ for $g \sim \mu$ and $\bar{U}_K := \frac{1}{K} \sum_{k=1}^K f(y | g_k \cdot x)$. If $\text{Var}(U) < \infty$ and $g_1, \dots, g_K \stackrel{\text{i.i.d.}}{\sim} \mu$, then $\text{Var}(\bar{U}_K) = \text{Var}(U)/K$ and $\text{Std}(\bar{U}_K) = \text{Std}(U)/\sqrt{K}$.*

F PROOF OF THEOREM 2

Let $\varphi_1, \dots, \varphi_K : \mathcal{X} \rightarrow \mathcal{V}$ be the view family and let $\sim$ be the equivalence relation on $\mathcal{V}$ generated by $\varphi_v(x) \sim \varphi_w(x)$ for all $x \in \mathcal{X}$ and all v, w . Write $q : \mathcal{V} \rightarrow \mathcal{V}/\sim$ for the quotient map and $Q := q(\varphi_1(X))$; by construction of $\sim$ we have $q(\varphi_v(X)) = Q$ for every v , so Q is well defined without reference to which presentation was received.

Step 1: orbit-constant predictors are exactly the Q -measurable ones. A predictor f on $\mathcal{V}$ satisfies $f(\varphi_v(x)) = f(\varphi_w(x))$ for all x, v, w if and only if f is constant on $\sim$ -classes, which is to say $f = \tilde{f} \circ q$ for some $\tilde{f}$. Minimising log loss over this class gives the conditional law of Y given Q , so

$$R_{\text{inv}} = \inf_{\tilde{f}} \mathbb{E}[-\log \tilde{f}(Y | Q)] = H(Y | Q).$$

Because f is orbit-constant the value does not depend on which v generates the input, so no mixing distribution over v need be specified.

Step 2: Q is a coarsening of every view. For each v , $Q = q(\varphi_v(X))$ is a measurable function of $\varphi_v(X)$, hence $\sigma(Q) \subseteq \sigma(\varphi_v(X))$. Conditioning on a coarser σ -algebra cannot decrease conditional entropy, so

$$R_{\text{inv}} = H(Y | Q) \geq H(Y | \varphi_v(X)) = R(\varphi_v) \quad \text{for every } v.$$

Step 3. Taking the maximum over v gives $R_{\text{inv}} \geq \max_v R(\varphi_v)$, and subtracting $\min_v R(\varphi_v)$ from both sides gives $R_{\text{inv}} - \min_v R(\varphi_v) \geq \delta$. $\square$

Equality, and the two certificates. If the views are mutually reconstructible, each $\varphi_v(x)$ determines every $\varphi_w(x)$, so each $\sim$ -class meets each view's range once and $\sigma(Q) = \sigma(\varphi_v(X))$. Step 2 is then an equality and $R_{\text{inv}} = R(\varphi_v)$: a bijective relabelling costs nothing. Reconstructibility is not needed for that conclusion. If $\varphi_w = c \circ \varphi_v$ for a coarsening c that is sufficient for Y , meaning $H(Y | \varphi_v(X)) = H(Y | c(\varphi_v(X)))$, then $\delta = 0$ and the bound is again tight at zero even though c is not invertible. Angle wrap is this case, which is why the test reports reconstructibility as a certificate and gates only on parity.

Beyond log loss. Only two properties of log loss were used: that its Bayes risk is the conditional entropy, and that this risk is concave in the predictive distribution so coarsening cannot decrease it. Both hold for the generalized entropy of any strictly proper scoring rule, so the statement carries over with H replaced by that rule's Bayes risk.

Algorithm 1 Audit and repair for a certified symmetry

Require: base model f_T , nuisance sampler μ , orbit samples K , metric M , dispersion statistic A , tolerance ε

- 1: Certify the transformation family (§3); abort unless predictive parity holds
- 2: Measure $A(f_T)$ and $M(f_T)$ on held-out data
- 3: Fit g_θ by Eq. (2) for probability- or direction-valued outputs
- 4: Bootstrap confidence intervals for ΔM and ΔA
- 5: **return** MAKE if $\text{CI}[\Delta A] < 0$ and $\text{CI}[\Delta M] \geq -\varepsilon$, else BREAK

G DIRECTION-VALUED PREDICTION HAS NO JENSEN GUARANTEE

Fix an input x and nuisance sampler μ . Given K samples $g_1, \dots, g_K \sim \mu$, let $\bar{p} := \frac{1}{K} \sum_{k=1}^K f(\cdot | g_k \cdot x)$. The empirical Jensen gain

$$\hat{J}(x; y) := \frac{1}{K} \sum_{k=1}^K (-\log f(y | g_k \cdot x)) - (-\log \bar{p}(y)) \geq 0$$

is non-negative by concavity of the logarithm. This is why orbit averaging is safe wherever the prediction is a probability, and it is the guarantee the setting is not one where that guarantee applies.

The guarantee needs the prediction to be a probability vector and the score to be log loss. Neither holds when the prediction is a unit vector scored by cosine similarity, and Proposition 3 is the counterexample. Take $d = 2$ and $y = (1, 0)$, with orbit predictions

$$u_1 = (1, 0), \quad u_{2,3} = \left(-\frac{9}{10}, \pm \sqrt{1 - \frac{81}{100}}\right),$$

all of unit norm. The individual scores are $\langle u_1, y \rangle = 1$ and $\langle u_2, y \rangle = \langle u_3, y \rangle = -\frac{9}{10}$. Their sum is $\sum_k u_k = (-\frac{8}{10}, 0)$, so $\bar{u} = (-1, 0)$ and $\langle \bar{u}, y \rangle = -1$, strictly below all three. The inequality $-1 < -\frac{9}{10}$ is strict, so the witness survives any sufficiently small perturbation of y and of the u_k and is not an artifact of exact symmetry. Taking instead $u_{1,2} = (0, \pm 1)$ gives $\sum_k u_k = 0$, where $\bar{u}$ is undefined; that configuration is not itself stable, but $\|\sum_k u_k\|$ is arbitrarily small in a neighbourhood of it, so the normalized mean is ill-conditioned there. Both failures are properties of the output space rather than of any particular predictor. They rule out averaging the outputs, not averaging as such, which is why Eq. (2) keeps the average and moves it to the loss.

Audit and repair. Orbit averaging supplies a natural teacher, and a student can be distilled to match it. We use that as the falsifiable make/break test of Definition 1: if distillation reduces dispersion without materially harming task performance, the nuisance family is a plausible invariance constraint, and otherwise it is rejected. Algorithm 1 states the procedure; for direction-valued predictions the distillation step is replaced by Eq. (2) for the reason above.

H SUPPLEMENTARY EXPERIMENTS**H.1** ACTION-CONDITIONED DYNAMICS: ROBOTICS AND iPhyRE

Three robotics tasks (ALOHA, LIBERO-10, PushT) and one physics simulation (iPhyRE). Frozen ViT-H/14 JEPa encoder; predictor receives (z_t, a_t) and predicts future latent. Evaluation by retrieval (lower mean rank is better). Key intervention: action-shuffle (deliberately not evidence-preserving).

Action conditioning produces large improvements on ALOHA and LIBERO-10; action-shuffle degrades performance below baseline, confirming causal use of the action signal. PushT yields BREAK: adding actions hurts retrieval rank ($179.9 \rightarrow 349.9$), indicating the action representation is incompatible with the encoder’s latent geometry. iPhyRE shows the clearest

Table 8: Action conditioning: action-shuffle degrades performance, confirming causal use.

Task	rank↓ (no-act)	rank↓ (act)	rank↓ (shuf)	R@10↑ (no-act)	R@10↑ (act)	Verdict
ALOHA	186.2	97.7	635.8	0.068	0.124	MAKE
LIBERO-10	80.9	49.6	55.5	0.136	0.242	MAKE
PushT	179.9	349.9	547.7	0.079	0.093	BREAK

separation: action conditioning improves mean rank $5.5\times$ ($172.6 \rightarrow 31.3$); shuffled actions revert to baseline (173.9).

Table 9: iPhyRE: $5.5\times$ retrieval improvement with action conditioning.

Metric	persist	no action	action	shuf
mean rank↓	167.5	172.6	31.3	173.9
Recall@10↑	0.203	0.185	0.471	0.124

H.2 MOLECULAR DYNAMICS: REPRESENTATION ABLATION

Lennard-Jones fluid with exact $SE(3)\times S_n$ symmetry. Two encoders: RDF (invariant) vs. flattened coordinates (not aligned). Same data, same training, same transformation family ($K = 16$).

The RDF encoder is $SE(3)$ -invariant by construction: orbit dispersion is already small (0.012–0.017) and distillation preserves accuracy ($0.894 \rightarrow 0.893$). The flattened-coordinate encoder breaks $SE(3)$; orbit-averaging over rotations and permutations reduces accuracy catastrophically ($0.856 \rightarrow 0.456$) because the representation is not aligned with the symmetry. Same physical system, same symmetry group, same training. The representation determines the verdict.

Table 10: Same symmetry, two representations. The defect is caused by the representation.

Encoder	Acc (base)	Acc (stud.)	TV-mean (base)	TV-mean (stud.)	Verdict
RDF ($\sigma=0.01$)	0.894	0.893	0.017	0.016	MAKE
RDF ($\sigma=0.03$)	0.894	0.893	0.012	0.011	MAKE
Flatten ($\sigma=0.01$)	0.856	0.456	0.449	0.433	BREAK